

Worked Examples

These examples give instructors and students concrete models of the evidence expected in the Studio Refresh. They are intentionally short enough to discuss in class.

Use them in three ways:

- Before a sprint, show the structure of a strong artefact.
- During peer review, compare student evidence against the example.
- After class, use the examples as feedback language.

Week 1 - Strong Agent Work Order

Goal:

Create a reproducible weekly KPI workflow for the COO.

Inputs:

- week1_customers.csv
- week1_sales.csv
- week1_support.csv
- week1_metric_definitions.md

Definitions:

- Paid active accounts: `subscription_status` is active and `is_paid` is true.
- Net June revenue: `sum gross_amount` minus `refund_amount` for June orders.
- Open support tickets: tickets with blank `closed_date`.

Acceptance criteria:

- Script runs from a clean folder.
- Paid active accounts equals 6.
- Net June revenue equals 2660.
- Open support tickets equals 3.
- Output includes a short notes section listing assumptions.

Constraints:

- Do not redefine active customer.
- Do not use recent login as a substitute for paid active account.
- If a definition is missing, stop and ask.

Why This Is Strong

- It gives the agent a job, not a vague wish.
- It names source files and approved definitions.

- It includes known checks that can fail.
- It tells the agent what not to do.

Weak Version

Make a KPI report and explain the numbers.

This is weak because the agent can choose definitions, output format, and checks without supervision.

Week 1 - Example AI-Use Log Entry

Task given to agent	Output accepted	Output rejected or changed	Evidence used
Draft a Python script for weekly KPI checks.	File loading structure and revenue calculation.	Agent counted active customers using last login instead of paid active subscription.	<code>week1_metric_definitions.md</code> and known check: paid active accounts equals 6.

Week 2 - Row-Count Bridge

Step	Row count	Notes
Subscriptions source	10	Billing system scope.
Product events source	9	Includes C999, not in billing system.
Left join subscriptions to product events	10	Keeps all subscription customers.
Customers without product events	1	C204 requires quarantine.
Product-only records	1	C999 is outside billing scope.

Example Quarantine Rows

customer_id	issue_type	why_it_matters	recommended_action
C204	Missing product event	Inner join would remove a cancelled subscription customer.	Keep in churn denominator; flag missing usage.
C999	Product event without subscription	Product activity is outside billing scope.	Exclude from billing churn; investigate source mapping.

Week 2 - Trusted-Number Note

We recommend using the billing churn number for the board paper because the board decision o

Week 3 - Leakage Audit

Column	Decision-time availability	Keep or exclude	Reason
tenure_months	Available	Keep	Known before offer decision.
monthly_fee	Available	Keep	Billing attribute.
usage_sessions_30d	Available	Keep	Observed before decision.
support_tickets_90d	Available	Keep	Observed before decision.
plan_type	Available	Keep	Customer attribute.
discount_last_90d	Available	Keep with caution	May reflect prior intervention.
post_offer_response	Not available	Exclude	It happens after offer decision.
retained_next_month	Not available	Target only	This is the outcome being predicted.

Week 3 - Decision Memo Excerpt

Recommendation:

Run a limited retention pilot for the top 30 percent of customers ranked by churn risk, excl

Evidence:

The candidate policy is compared against two baselines: highest support burden and lowest us

Decision logic:

At an offer cost of 25 and expected saved margin of 120, the policy needs to save more than

Limit:

This is not a causal estimate of the offer effect. It is a prioritisation policy for a pilot

Week 4 - Input Contract

```
{
  "customer_id": "string, required",
  "tenure_months": "integer >= 0, required",
  "monthly_fee": "number >= 0, required",
  "usage_sessions_30d": "integer >= 0, required",
  "support_tickets_90d": "integer >= 0, required",
  "plan_type": "one of: monthly, annual, enterprise"
}
```

Expected Responses

Request	Expected behaviour
Valid request for C301	Return recommendation, score or rule, and version.
Missing customer_id	Reject with clear error message.
Negative tenure_months	Reject with clear error message.
Unknown plan_type	Reject with allowed values.

Week 4 - Runbook Excerpt

Health check:

Run the good-input request and confirm the response includes recommendation, reason, and ver

Known failure:

Malformed request causes validation failure.

Immediate action:

Do not edit production data. Capture request payload, confirm validation message, and route

Rollback:

Use previous rule version v0.1 if current version rejects valid requests.

Week 5 - Safe Executive Answer

June churn rose from 9.4 percent to 11.8 percent. The increase is concentrated among monthly

One plausible explanation is onboarding friction: the Support Summary says onboarding-delay

The evidence does not support saying the May price increase caused the churn increase, beca

Recommended next test:

Compare churn among low-usage new customers with and without onboarding-delay tickets, then

Why This Is Strong

- It separates observation from explanation.
- It cites evidence that supports each claim.
- It refuses an unsupported causal claim.
- It proposes a next test.

Week 6 - Launch Recommendation

Recommendation:

Limited pilot, not full launch.

Scope:

Use the automation for weekly KPI preparation and internal review. Do not use the analyst-ag

Strongest evidence:

The KPI workflow has smoke checks for paid active accounts, net revenue, and open support t

Residual risk:

The retention decision model has been checked for obvious leakage, but the business impact o

Owner:

Analytics lead owns monitoring and rollback. Customer Success owns review of quarantined car

Feedback Language For Instructors

Use feedback that points to evidence:

- “Your artefact runs, but your defence does not name the failure it prevents.”
- “This is a useful agent prompt, but it needs an acceptance test.”
- “The model score is not yet a decision policy.”

- “The answer is fluent, but the cited source does not support that causal claim.”
- “A limited pilot is a valid launch recommendation because you named scope and owner.”